

Max seems to talk of the coherence theorem for monoidal categories as a kind of normalization theorem.

In some sense I see the idea.

We can build a normalizer for the free category over a signature if we can define a separate category with a weak initiality property

Then we argue that

$$F[\Sigma] \xrightarrow{\text{rec}_F(L)} C \\ \xleftarrow{\text{rec}_C(\tau)}$$

$L: \Sigma \rightsquigarrow C$
and τ the tautological interpretation
From which

We get $\text{rec}_F(L) \circ \text{rec}_C(\tau)$
This round trip shouldn't quite
be the identity but it
should be an idempotent?

In the proof of the coherence theorem, max instead argues that this composite is the identity, then because the category of lists is thin, this retract shows that the free monoidal category is thin

Can a similar setup be used for a normalization proof?